

JOVIAL JOYSON

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SUMMARY

EEE undergraduate interested in **power electronics, EV systems, embedded control, and industrial automation**. Experience in BMS, PLC automation, IoT systems, embedded hardware integration, and MATLAB-based simulation. Co-author of an Indian patent application on intelligent electrical motor monitoring systems.

EDUCATION

B.Tech – Electrical & Electronics Engineering, Christ College of Engineering Expected May 2027

CGPA: 7.83 / 10

Don Bosco HSS

12th: 92% 10th: 95%

PROJECTS

EBAJA EV Competition

2025

- Contributed to Battery Management System (BMS) design for monitoring voltage and temperature parameters of EV battery cells.
- Assisted in ECU interfacing, subsystem wiring, and embedded sensor integration for vehicle electronics.
- Performed MATLAB-based simulations for battery behavior analysis and system validation.

PLC Pharmacy Automation

- Engineered an automated medicine dispensing system using Siemens PLC and HMI interfaces.
- Programmed ladder logic for sequential actuator control and fault-safe operation.
- Streamlined monitoring and dispensing workflow through industrial automation architecture.

Wireless Robotic Arm

- Built a multi-axis ESP32-based robotic arm with wireless remote control capability.
- Implemented motor driver interfacing for directional and precision movement control.
- Enabled wireless manipulation through embedded communication architecture.

Smart Home Automation

- Designed an ESP32-based IoT system for wireless appliance monitoring and control.
- Incorporated sensors and relay modules for real-time switching operations.

Animal Deterrent System

- Created an ML-based real-time animal detection system using computer vision concepts.
- Triggered automated deterrent mechanisms based on live detection output.

EXPERIENCE

Industrial Automation Intern – Siemens COE, Anna University

Jun – Jul 2025

- Programmed industrial PLC workflows using Siemens TIA Portal and ladder logic.
- Designed SCADA and HMI interfaces for process monitoring and visualization.
- Improved automation reliability through structured control and monitoring architecture.

Client Management Intern – White Matrix Software Solutions

Jan 2026 – Present

- Coordinated client requirements and translated them into structured technical documentation.
- Facilitated communication between development teams and external stakeholders.

PATENTS

Context-Aware Digital Triplet System for Electrical Motors Under Variable Load

2026

- Indian Patent Application Filed (Application No: 202641051670)
- Contributed to embedded sensing and predictive motor-state monitoring concepts under varying load conditions.

ACHIEVEMENTS

- Winner & Best Pitch – BeachHack H4C Hackathon
- 1st Place – Pitch Perfect Idea Competition
- Finalist – Beach Hack AI Challenge

SKILLS

Technical: PLC (TIA Portal), SCADA, HMI, ESP32, Arduino, IoT, MATLAB, AutoCAD

Core Areas: Power Electronics, Electric Vehicles, Embedded Systems, Energy Systems

Programming: Python, C++, Java (Basic)

Certifications: IIT Madras Pravartak (Intelligent Data Center O&M), NPTEL – Introduction to Machine Learning

Leadership: Communication Head – Meta Hub

Languages: English, Malayalam, Hindi